I am submitting under Pathway A. My project is based on: AXK Network: A Verification-and-Settlement Layer for Cross-Border Agricultural Trade. Every screenshot, design file, arguments in this document is real and is my own original work.

Axk

Verified Commodity Settlement on Stellar

AXK links people, commodity locations, inspection decisions and payments.

**This audit tests who that design protects, and who it could leave behind.**

ETHICS IN SOFTWARE ENGINEERING · SUMMATIVE ASSESSMENT · PATHWAY A

# Ethics in Practice

An ethical, legal and societal audit of the AXK Network Capstone verification-and-settlement system

Video defense (Google Drive)

Link to be added


Prepared by

Oghenetejiri Jesse

Institution

African Leadership University

Academic supervisor

Junior Turatsinze

Course

Ethics in Software Engineering

REC approval

SEPT26-BSE-039, letter dated 2 November 2026

Contact

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AT A GLANCE

# AXK makes trade decisions visible. **This audit shows where that protects people, and where it does not yet.**

In the Capstone, cooperatives register lots, independent inspectors record pass or fail decisions, buyers place test tokens into escrow, and a public page links transactions and record digests. It runs on the Stellar testnet and moves no real money. This audit reads that system against three ethical risks and five legal and professional frameworks, using evidence pinned to one commit.

## WHERE THE RISK SITS

### Trade and participant data

Full records stay off-chain, but grades, amounts and addresses are on-chain, and the public page shows vehicle plates and customs details.

**Digests on-chain and a fixed public field list.**

### Inspection decisions

A failed inspection blocks payment, and it cannot be corrected or appealed.

**Separation of duties holds for keys, not yet for users.**

### Access and inclusion

Verification can favour organisations with stable internet, devices and English.

**ODK offline capture helps inspectors. No translations yet.**

### Privacy and legal duties

/privacy returned 404 on the live site on 4 November 2026.

**Policy page prepared and tested locally, not yet deployed.**

**3**
ethical risks, with the evidence inside each risk table

**5**
frameworks: local, regional and international

**16**
findings in the technical evidence register

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**Table 1**

*Submission Details*

| Submission field | Details |
|---|---|
| **Pathway** | A: Mission Capstone |
| **Student** | Oghenetejiri Jesse |
| **Academic supervisor** | Junior Turatsinze |
| **REC approval** | SEPT26-BSE-039; letter dated 2 November 2026 |
| **Repository** | https://github.com/tejirijesse/axk-open-source-capstone (evidence pinned to commit `44629ba`) |

# Project and Evidence Boundary

AXK Network links verification of agricultural commodity lots with settlement of cross-border trades. In the Capstone, cooperatives register lots, independent inspectors record pass or fail decisions, buyers place test tokens into escrow, logistics and port steps are recorded, and a public page links transactions and record digests. The system combines a Frappe and MariaDB application with four Soroban smart contracts on the Stellar testnet. These are testnet demonstrations; they do not prove readiness for live financial services or real commodity settlement (Jesse, 2026a, 2026b).

The public Capstone repository is the technical evidence base for this audit. Unless stated otherwise, code figures show commit `44629ba` and system figures were captured on 4 November 2026.

# Part One: Research Ethics Approval

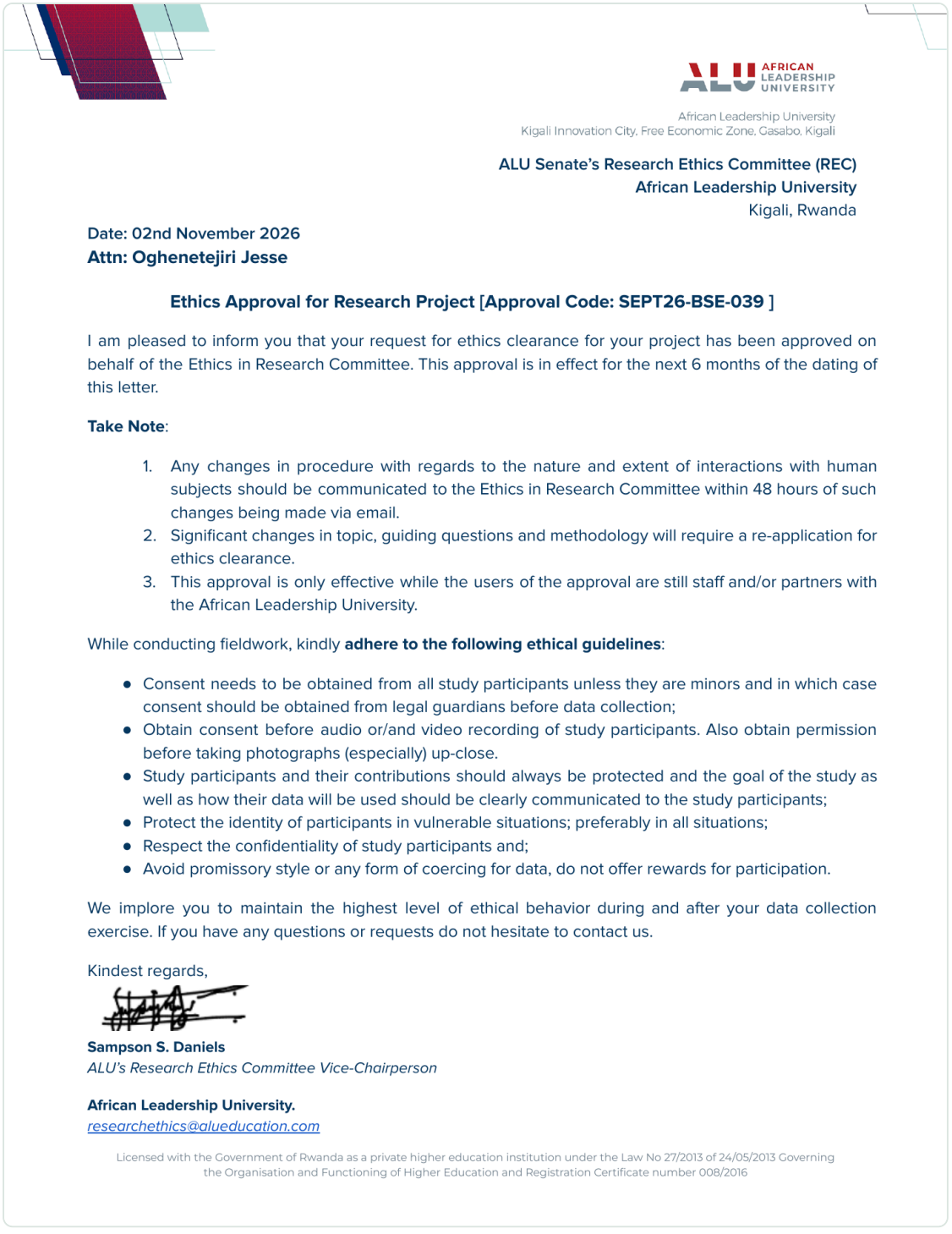
## 1.1 Formal Approval and Participant Consent

I received an ALU Senate Research Ethics Committee approval letter dated 2 November 2026, approval code SEPT26-BSE-039. Approval runs for six months from that date. The letter requires consent from all participants, separate consent for recording, protection of identity and confidentiality, no rewards for participation, notification of procedural changes within 48 hours, and re-application for significant changes (African Leadership University Senate Research Ethics Committee, 2026).

The study consent form describes a session of about one hour: a prototype role exercise, a questionnaire and an optional conversation. It states that testnet transactions involve no real money or produce, and it sets out voluntary participation, a withdrawal route, anonymised reporting and separate recording permission (Jesse, 2026c).

**Figure 1**

*ALU Research Ethics Committee Approval Letter*



African Leadership University
Kigali Innovation City, Free Economic Zone, Gasabo, Kigali

**ALU Senate's Research Ethics Committee (REC)**
**African Leadership University**
Kigali, Rwanda

**Date: 02nd November 2026**
**Attn: Oghenetejiri Jesse**

**Ethics Approval for Research Project [Approval Code: SEPT26-BSE-039 ]**

I am pleased to inform you that your request for ethics clearance for your project has been approved on behalf of the Ethics in Research Committee. This approval is in effect for the next 6 months of the dating of this letter.

**Take Note**:

1. Any changes in procedure with regards to the nature and extent of interactions with human subjects should be communicated to the Ethics in Research Committee within 48 hours of such changes being made via email.
2. Significant changes in topic, guiding questions and methodology will require a re-application for ethics clearance.
3. This approval is only effective while the users of the approval are still staff and/or partners with the African Leadership University.

While conducting fieldwork, kindly **adhere to the following ethical guidelines**:

- Consent needs to be obtained from all study participants unless they are minors and in which case consent should be obtained from legal guardians before data collection;
- Obtain consent before audio or/and video recording of study participants. Also obtain permission before taking photographs (especially) up-close.
- Study participants and their contributions should always be protected and the goal of the study as well as how their data will be used should be clearly communicated to the study participants;
- Protect the identity of participants in vulnerable situations; preferably in all situations;
- Respect the confidentiality of study participants and;
- Avoid promissory style or any form of coercing for data, do not offer rewards for participation.

We implore you to maintain the highest level of ethical behavior during and after your data collection exercise. If you have any questions or requests do not hesitate to contact us.

Kindest regards,

**Sampson S. Daniels**
*ALU's Research Ethics Committee Vice-Chairperson*

**African Leadership University.**
researchethics@alueducation.com

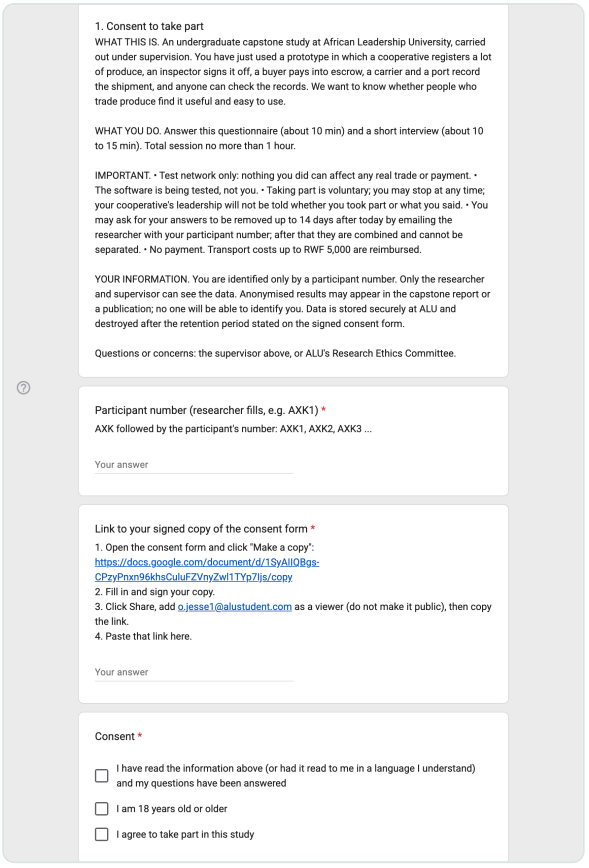
Licensed with the Government of Rwanda as a private higher education institution under the Law No 27/2013 of 24/05/2013 Governing the Organisation and Functioning of Higher Education and Registration Certificate number 008/2016

*Note.* ALU Senate Research Ethics Committee approval letter, approval code SEPT26-BSE-039, dated 2 November 2026.

Twenty participants took part on 8 November 2026, six days after approval. All 20 ticked the three consent statements in the questionnaire before answering, and pasted a link to a signed copy of the consent form. All 20 declined an interview recording, so no recordings were made.

**Figure 2**

*Consent Section of the Post-Use Evaluation Form*




*Note.* Consent section of the AXK Network post-use evaluation form, captured from the live form on 11 November 2026. Participants read the study information and ticked all three statements before any question was shown.

## 1.2 Approval Scope and Chronology

The approval letter is dated 2 November 2026. An earlier consent form I hold records a session dated 1 October 2026, and some evaluation records carry dates in October, before approval. The letter does not state that it covers earlier activity, so I have not used any of those records: they are excluded from this audit and from the capstone report. All participant data I use was collected on 8 November 2026, after approval.

The letter also says not to offer rewards for participation. The consent form offers reimbursement of transport costs of up to RWF 5,000, but participants were nearby or reached by phone, and no one was paid or reimbursed.

My obligations extend beyond a committee letter. I must protect participant identity, explain how data will be used, restrict access to identifiable material, and keep simulated transactions separate from real commerce. Any move from a supervised prototype to a commercial rollout needs its own legal and ethical assessment.

**I have checked the ethical risks of my project. I got formal ethical approval from the right authority before collecting data, testing with users, or launching the software.**

# Part Two: Ethical Risks and System Proof

I selected risks by the severity of harm, the exposure of vulnerable stakeholders, reversibility, and the trust a trade-and-settlement tool requires. I applied a rights-based view (privacy, fairness, non-discrimination), consequentialist reasoning (how failures affect users) and professional duties to minimise harm (Association for Computing Machinery, 2018).

## Cross-risk Ethical Decision Record

**Table 2**

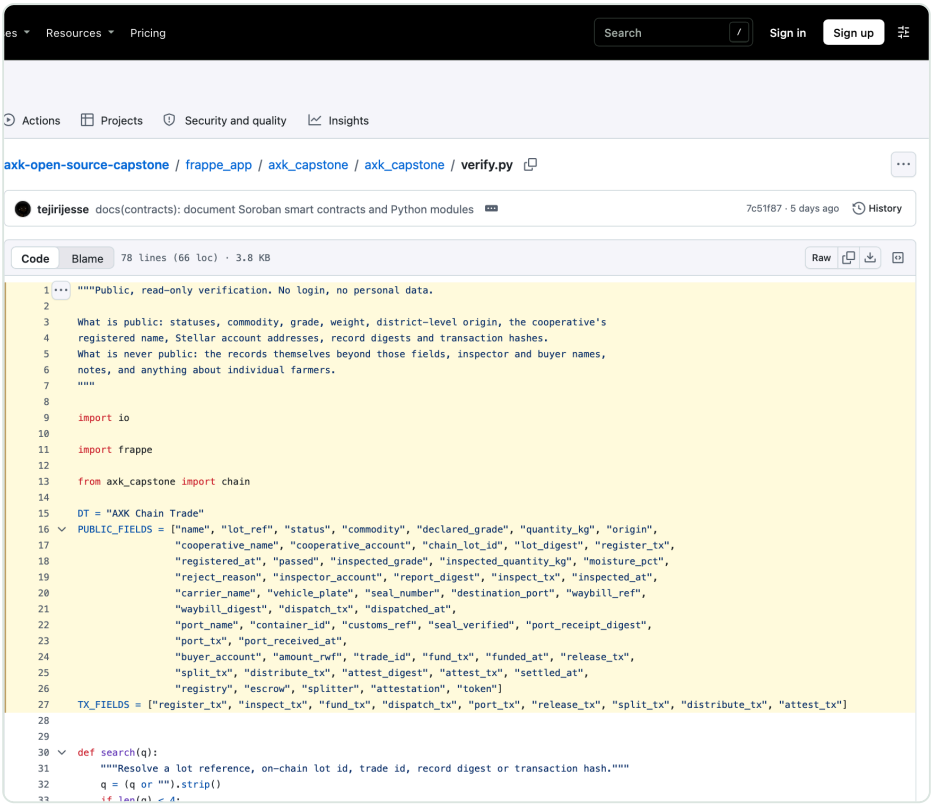
*Cross-risk Ethical Decision Record*

| Decision | Value gained | Trade-off or residual risk |
|---|---|---|
| **Keep full records off-chain** | Reduces unnecessary public disclosure | Grades, quantities, amounts, hashes and addresses remain public and may be linkable |
| **Independent, write-once inspection** | Reduces self-dealing and fraud | Mistaken or biased judgments are permanent, with no appeal yet |
| **Offline ODK inspection** | Field capture without live connectivity | Upload, training and accessibility gaps remain |
| **Server-held testnet keys** | Simple supervised prototype testing | Users must trust the operator; not production-grade custody |
| **Shared one-click demo logins** | Graders and testers can try every role at once | Signed-in screens are effectively public, so no real data may be entered |

**Table 3**

*Risk 1: Exposure and Misuse of Trade or Participant Data*

| Aspect | Analysis and evidence |
|---|---|
| **Risk explanation** | AXK links people, commodity locations, inspection decisions and payments. Exposure could enable fraud, reputational harm or commercial surveillance, and a public blockchain cannot be corrected after publication. The Capstone keeps full records off-chain, but inspection grades, quantities, amounts, addresses and digests are on-chain, and the public page shows carrier, vehicle plate and customs details. Shared demo logins also expose signed-in screens. Hashes and addresses may still be linkable. |
| **Who is excluded or disadvantaged** | People who cannot safely disclose identifying information, including vulnerable producers, may avoid onboarding. A public record that is safe for one cooperative can create reputational or security risk for another, and carriers can be identified through a vehicle plate. |
| **Design decision and alternatives** | I chose off-chain records with SHA-256 digests over placing full records on an immutable ledger, and a fixed list of public fields in `verify.py` that leaves out driver, inspector and buyer names and inspection notes. Rejected alternative: publishing full records on-chain for maximum transparency, which would make personal data permanent. |

| Aspect | Analysis and evidence |
|---|---|
| Status and remaining limits | **Implemented:** digest-only records, fixed public field list. **Not implemented:** retention and deletion workflow, re-identification assessment, per-person accounts on the hosted site, managed key storage. Grade, quantity and amounts remain readable on-chain by design, because the contracts need them. |
| Evidence | **Figure 3**<br>*Fixed List of Public Fields*<br>  <br>*Note.* `frappe_app/axk_capstone/axk_capstone/verify.py`, lines 1 to 27, commit 44629ba. The fixed list of public fields; names of the driver, inspector and buyer are absent. |



```
"""Public, read-only verification. No login, no personal data.

What is public: statuses, commodity, grade, weight, district-level origin, the cooperative's
registered name, Stellar account addresses, record digests and transaction hashes.
What is never public: the records themselves beyond those fields, inspector and buyer names,
notes, and anything about individual farmers.
"""

import io

import frappe

from axk_capstone import chain

DT = "AXK Chain Trade"
PUBLIC_FIELDS = ["name", "lot_ref", "status", "commodity", "declared_grade", "quantity_kg", "origin",
                 "cooperative_name", "cooperative_account", "chain_lot_id", "lot_digest", "register_tx",
                 "registered_at", "passed", "inspected_grade", "inspected_quantity_kg", "moisture_pct",
                 "reject_reason", "inspector_account", "report_digest", "inspect_tx", "inspected_at",
                 "carrier_name", "vehicle_plate", "seal_number", "destination_port", "waybill_ref",
                 "waybill_digest", "dispatch_tx", "dispatched_at",
                 "port_name", "container_id", "customs_ref", "seal_verified", "port_receipt_digest",
                 "port_tx", "port_received_at",
                 "buyer_account", "amount_rwf", "trade_id", "fund_tx", "funded_at", "release_tx",
                 "split_tx", "distribute_tx", "attest_digest", "attest_tx", "settled_at",
                 "registry", "escrow", "splitter", "attestation", "token"]
TX_FIELDS = ["register_tx", "inspect_tx", "fund_tx", "dispatch_tx", "port_tx", "release_tx", "split_tx", "distribute_tx", "attest_tx"]


def search(q):
    """Resolve a lot reference, on-chain lot id, trade id, record digest or transaction hash."""
    q = (q or "").strip()
    if len(q) < 4:
```

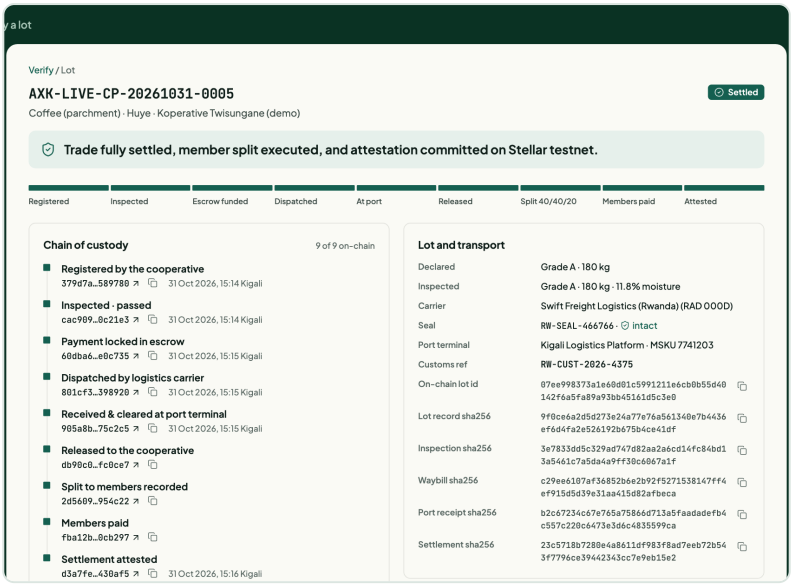
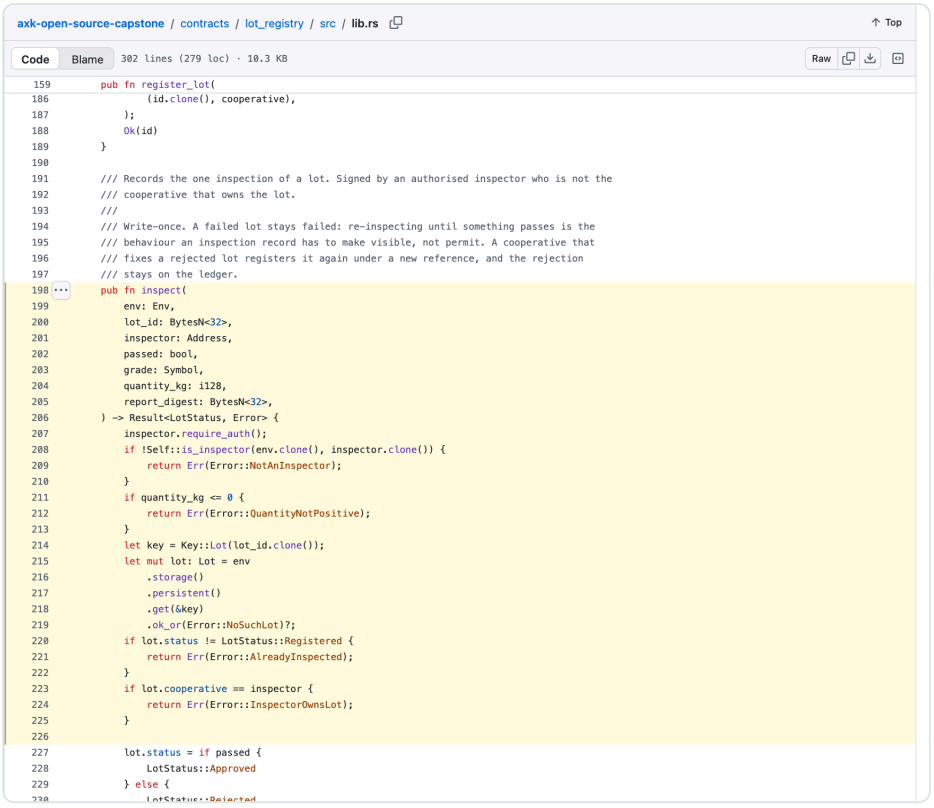
| Aspect | Analysis and evidence |
|---|---|
| Evidence | **Figure 4**<br>*Public Lot Page Without Sign-In*<br>  <br>*Note.* Public lot page on capstone.axk.org, 4 November 2026, no sign-in. It shows the carrier, vehicle plate and customs details, which is why the exposure is disclosed here. |

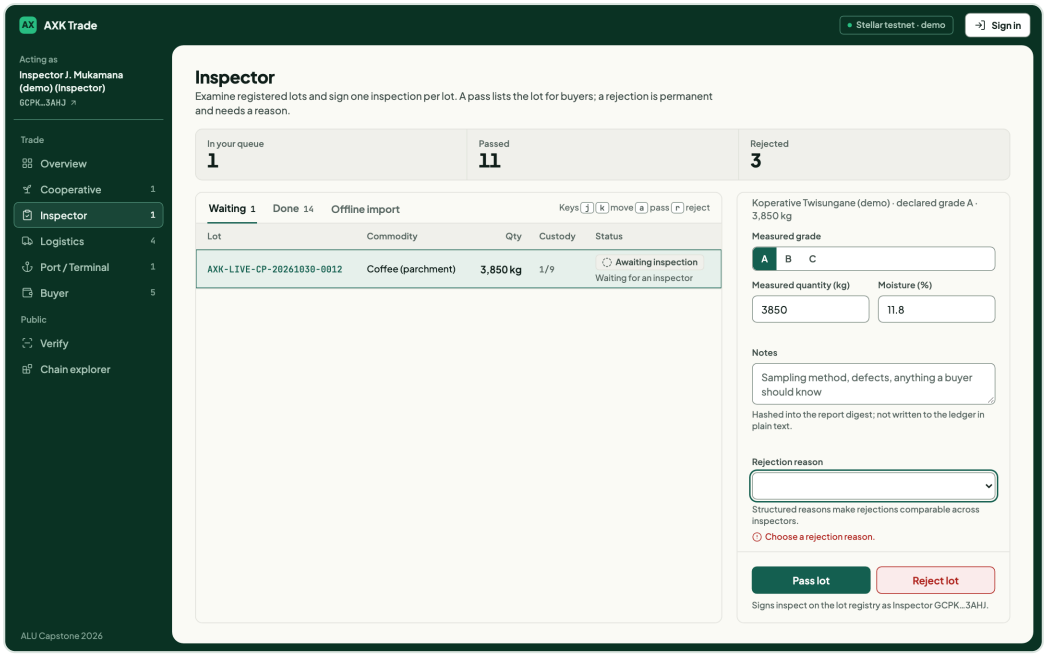


**Table 4**

*Risk 2: Unfair Inspection Decisions and Concentrated Authority*

| Aspect | Analysis and evidence |
|---|---|
| **Risk explanation** | A failed inspection blocks a cooperative from payment even if the inspector erred or acted improperly, and inspection is write-once, so no correction or appeal exists. The contract refuses a cooperative's own address as inspector and the server requires a listed rejection reason. However, the server holds every role's key and any signed-in demo login can trigger any role's action, so separation of duties is enforced for keys, not yet for users. |
| **Who is excluded or disadvantaged** | Small cooperatives that cannot reach a recognised inspector or challenge a disputed grade may lose access to buyer funding. People who do not understand a rejection reason are disadvantaged even when it is recorded. |
| **Design decision and alternatives** | I chose separation of duties, a fixed list of six rejection reasons and an inspection-gated escrow over self-certification, which would enable conflicts of interest. I made inspection write-once so a failed lot cannot be quietly re-inspected into a pass, accepting that this also makes honest mistakes permanent. |
| **Status and remaining limits** | **Implemented:** contract checks at `lot_registry` lines 208 to 225; server rejection-reason check at `trade_api.py` line 226; test `a_cooperative_cannot_inspect_its_own_lot` passes. **Not implemented:** an appeal or correction path; per-role checks in `_guard()` (lines 55 to 63), which only confirm the shared demo role; independent signing by carriers and ports (roadmap LOG-4, PORT-1). Buyer disputes are roadmap item BUY-4, not built. |

| Aspect | Analysis and evidence |
| --- | --- |
| Evidence | **Figure 5**<br>*Inspection Rules in the Lot Registry Contract*<br>  <br>*Note.* `contracts/lot_registry/src/lib.rs`, lines 198 to 226, commit 44629ba. Allow-listed inspector, one inspection per lot, and a refusal when the inspector owns the lot. |

| Aspect | Analysis and evidence |
|---|---|
| Evidence | **Figure 6**<br>*Rejection Refused Until a Reason Is Chosen*<br>  <br><br>*Note.* Inspector screen on the local full stack, 4 November 2026. Rejection is refused until a reason is chosen. |

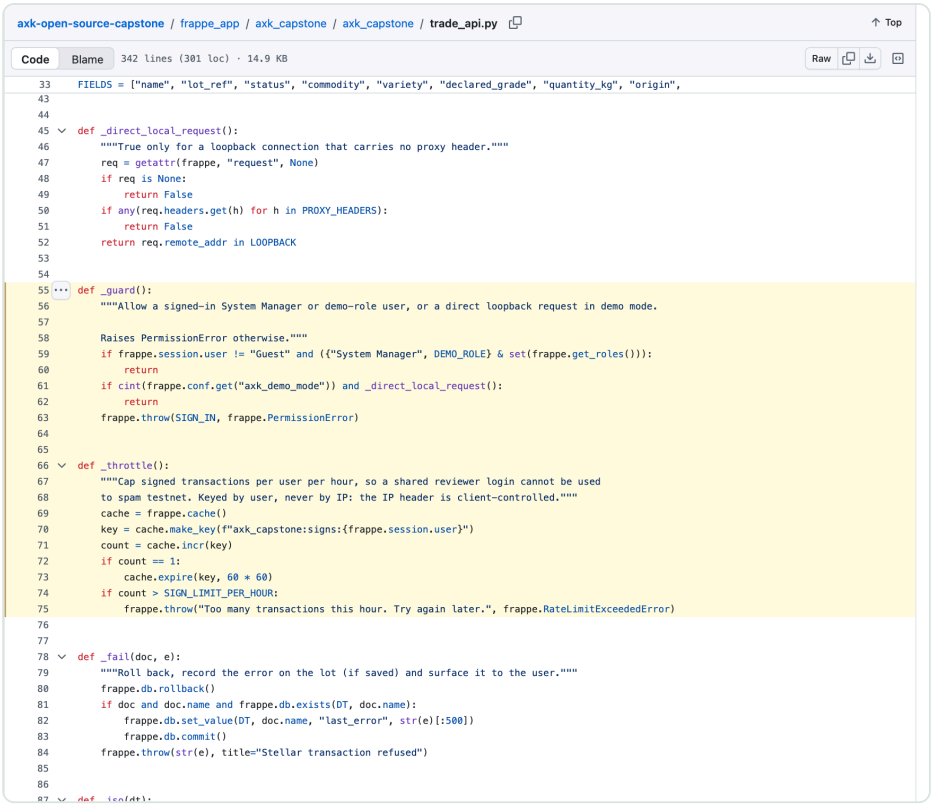
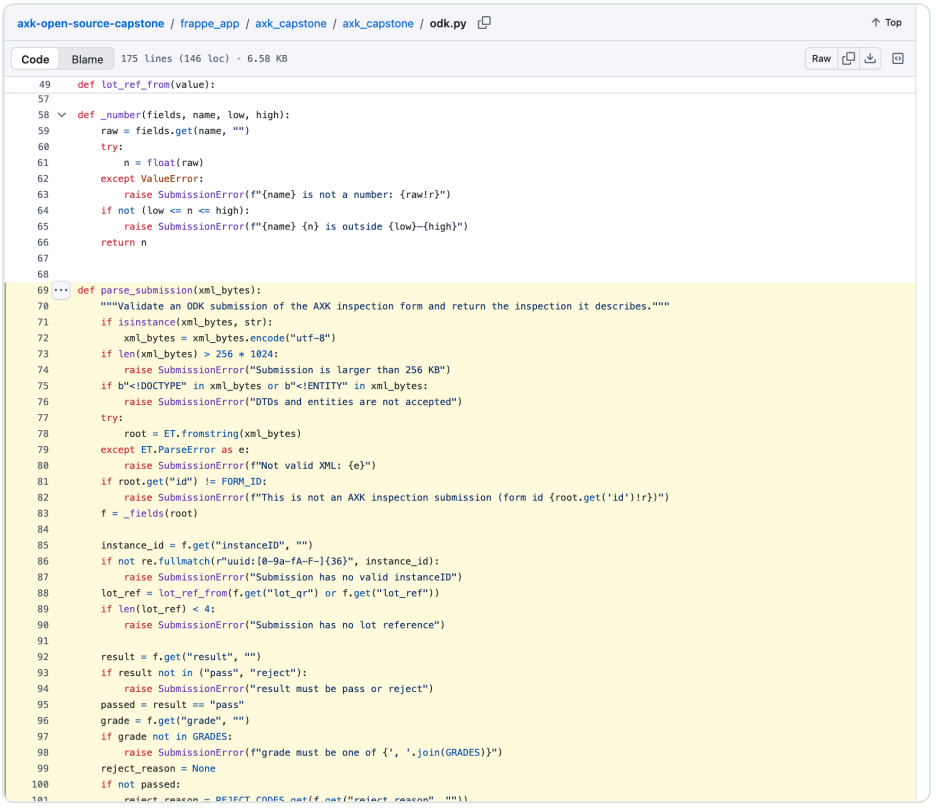
| Aspect | Analysis and evidence |
|---|---|
| Evidence | **Figure 7**<br>*Shared Role Check in the Server Guard*<br>  <br>*Note.* `trade_api.py` , lines 55 to 75, commit 44629ba. `_guard()` checks a shared role, not the specific role for each action. This is shown as a limit, not as a safeguard. |

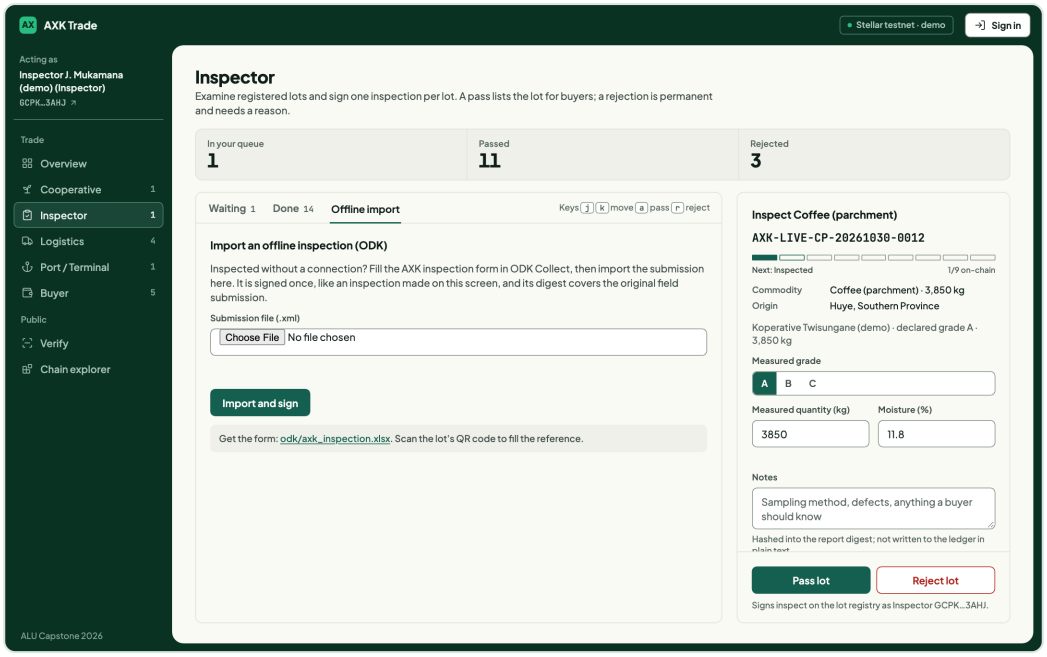
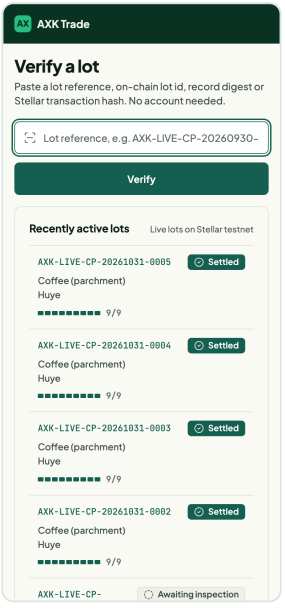


**Table 5**

*Risk 3: Connectivity, Device and Literacy Exclusion*

| Aspect | Analysis and evidence |
|---|---|
| **Risk explanation** | Farmers, inspectors and small cooperatives may have unstable connectivity, costly data or limited digital literacy. If verification needs uninterrupted internet and complex English screens, only better-resourced organisations can trade. The Capstone provides ODK offline inspection capture and was tested on a simulated slow-3G phone. Offline capture helps inspectors, but does not show that farmers can independently register lots, understand contract terms or use the system in their own language. |
| **Who is excluded or disadvantaged** | Users without smartphones, users with low literacy or visual disabilities, speakers of Kinyarwanda, French or Swahili, and anyone without an organisation or inspector able to upload offline records. The ODK path helps inspectors more than farmers. |
| **Design decision and alternatives** | I chose an offline capture path for inspections and small pages with no frontend framework, instead of assuming continuous connectivity. The importer checks form identity, values and rejection reasons before accepting a submission. Rejected alternative: a network-only inspection process. |

| Aspect | Analysis and evidence |
| --- | --- |
| Status and remaining limits | **Implemented and measured:** ODK import; 8 of 8 screens on a simulated phone (4x slower CPU, slow 3G), slowest 3.8 s ( `docs/testing/compatibility-local.md` ). **Not implemented:** translations (roadmap I18N-1), assisted onboarding, accessibility testing with affected users. Page speed is not evidence that people can complete their tasks. |
| Evidence | **Figure 8**<br>*Validation of an Offline Inspection Before Import*<br>  <br>*Note.* `frappe_app/axk_capstone/axk_capstone/odk.py` , lines 69 to 101, commit 44629ba. Validation of an offline inspection before import. |

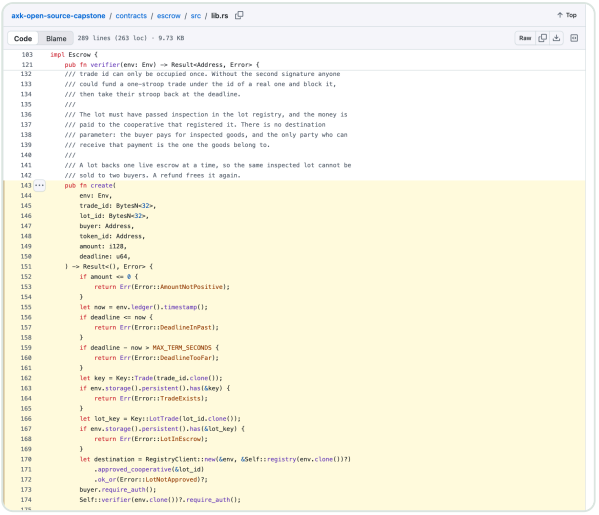
| Aspect | Analysis and evidence |
|---|---|
| Evidence | **Figure 9**<br>*Offline Inspection Import Screen*<br>  <br><br>*Note.* Offline import screen on the local full stack, 4 November 2026. |
| Evidence | **Figure 10**<br>*Public Verification Page at Phone Width*<br>  <br><br>*Note.* Public verification page on capstone.axk.org at phone width (390 px), 4 November 2026. |

## Two Safeguards to Demonstrate in the Video

**Safeguard 1: inspection-gated escrow with no payee argument.** The escrow asks the lot registry who owns an approved lot before taking any money, and refuses the call otherwise. Nobody can name a different payee. Tests `an_uninspected_lot_cannot_be_funded` and `the_payee_is_the_cooperative_that_registered_the_lot` passed when run on commit 44629ba.

**Figure 11**

*Inspection-Gated Escrow Funding*

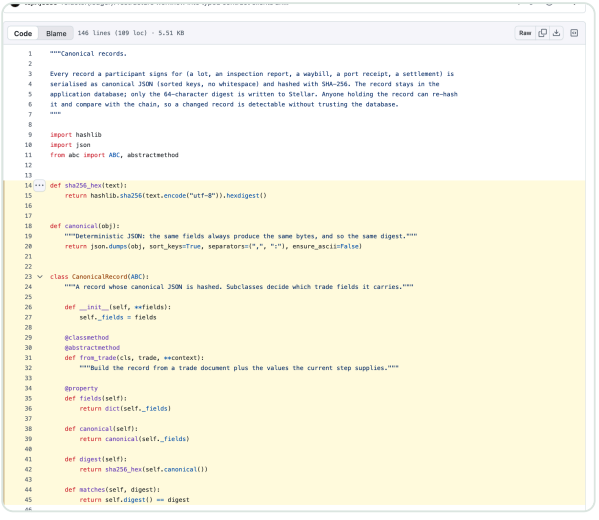



*Note.* `contracts/escrow/src/lib.rs`, lines 143 to 175, commit 44629ba.

**Safeguard 2: records off-chain, digests on-chain.** Each record is written as canonical JSON and hashed, and only the digest is anchored, so a changed record is detectable without publishing the record.

**Figure 12**

*Canonical Record Hashing*




*Note.* `frappe_app/axk_capstone/axk_capstone/ledger/records.py`, lines 14 to 45, commit 44629ba.

# Part Three: Laws and Video Defense

## 3.1 Local, Regional and International Frameworks

This is a feature-to-rule analysis, not a claim that AXK is legally compliant. Whether an obligation applies depends on establishment, users, processing location and other scope tests.

**Table 6**

*Legal and Regulatory Frameworks Matched to AXK Features*

| Instrument and provision | Rule and AXK feature | Status and next work |
|---|---|---|
| **Rwanda, Law No. 058/2021** (local; binding once real personal data is processed)<br>Arts. 37, 42, 23, 24, 47, 50 (Republic of Rwanda, 2021) | Art. 37: lawfulness, purpose limitation, accuracy, storage limitation. Art. 42: tell people who collects data, why, the recipients, retention and any transfer abroad. Arts. 23 and 24: erasure and rectification. Art. 47: security safeguards. Art. 50: store data in Rwanda unless certified to store abroad.<br>AXK feature: consent form, Privacy Policy, off-chain records, server-held keys. | Consent form and policy address Art. 42 in part. Immutable on-chain entries sit awkwardly with Arts. 23, 24 and 37 accuracy. Where the server database is hosted has not been confirmed, so Art. 50 compliance is not yet evidenced. Registration (Arts. 29 to 31) and any high-risk impact assessment (Art. 38) are not yet evidenced. |
| **Uganda, Data Protection and Privacy Act 2019** (local to the corridor; binding only once operating there)<br>ss. 7, 20 (Republic of Uganda, 2019) | s. 7: prior consent, with listed exceptions. s. 20: appropriate technical and organisational safeguards, including risk identification, verification and updates.<br>AXK feature: Rwanda to Uganda corridor; role sign-in, rate limit, testnet keys. | No Uganda deployment or Ugandan participant data is established, so this is forward planning. Per-role authorisation and breach response would be needed first. |
| **African Union, Malabo Convention (2014)** (regional; in force 8 June 2023)<br>Arts. 13, 19, 21 (African Union, 2014) | Art. 13: data processing principles, including consent, purpose and accuracy. Art. 19: rectification and erasure. Art. 21: security obligations.<br>AXK feature: cross-border design and digest-only records. | Binds ratifying states, and I read it as reaching AXK through national law. Rwanda and the DRC have ratified; Uganda and Kenya have not, so it is not corridor-wide. Supporting context, after Law 058/2021. |
| **EU GDPR, Regulation 2016/679** (international; benchmark)<br>Arts. 5(1)(c), 17, 25, 32; Recital 26 (European Parliament & Council of the European Union, 2016) | Art. 5(1)(c): data limited to what is necessary. Art. 25: data protection by design and by default. Art. 32: security appropriate to the risk. Art. 17: erasure. Recital 26: pseudonymised data can remain personal data.<br>AXK feature: fixed public field list, digests, on-chain addresses. | On the stated facts, Art. 3 does not make it apply, so it is a design benchmark. Recital 26 is why digests and addresses are not treated here as automatically anonymous. No erasure route exists for on-chain entries. |
| **ACM Code of Ethics (2018)** (international; professional code, not law)<br>ss. 1.2, 1.3, 1.4, 1.5, 1.6, 2.5, 2.9 (Association for Computing Machinery, 2018) | Avoid harm; be honest about limits; be fair; credit others' work; respect privacy; evaluate risks fully; build robustly and usably secure systems.<br>AXK feature: disclosed testnet limits, licence inventory, independent inspection. | Supports honest disclosure of the appeal gap (1.4), shared demo logins and server-held keys (2.9), and the third-party inventory (1.5). |

**Why these five.** Kenya's Data Protection Act (Republic of Kenya, 2019) was considered but left out: its s. 4 requires a Kenyan establishment or residence, or data subjects in Kenya, and the Capstone's corridors are Rwanda to the DRC and Uganda. The East African Community has no binding data protection instrument that I could find; its 2010 cyberlaws framework is guidance that does not set data protection rules (see the draft in East African Community & United Nations Conference on Trade and Development, 2008). ISO/IEC 27001:2022 (International Organization for Standardization & International Electrotechnical Commission, 2022) is a voluntary standard and was not used because its text is paywalled and could not be checked clause by clause.

**Limits.** Binding status is my reading, not legal advice. Whether synthetic demo data falls outside these laws, and whether Stellar addresses count as online identifiers under Rwanda's Art. 3, are arguable. Later amendments and Rwanda's implementing regulations were not exhaustively checked.

## 3.2 Intellectual Property and Third-Party Components

The Capstone repository is MIT licensed, with AFRIKABAL Inc. named as copyright holder in `LICENSE`. The escrow, splitter and attestation contracts originate in AXK Network's `AXKNETWORK/axk-stellar` repository (MIT), and the escrow was rebuilt for this Capstone. The screen redesign and hosted deployment included work contributed by Joseph Rukundo of AXK Network.

**Table 7**
*Code and Intellectual Property List*

| Component (version checked) | Role | Licence and how it was verified |
|---|---|---|
| **Frappe Framework 15.104.0** | Application server | MIT (LICENSE file in installed copy) |
| **soroban-sdk 22.0.11, soroban-env-host 22.1.3** | Contract framework | Apache-2.0 ( `cargo metadata` , Cargo.lock) |
| **194 Rust crates in Cargo.lock** | Contract dependencies | All permissive (MIT, Apache-2.0, BSD and similar); no GPL or missing licence fields ( `cargo metadata` ) |
| **stellar-cli 28.0.0** | Build, deploy, signing | Apache-2.0 (GitHub repository licence) |
| **stellar-sdk 13.2.1 (Python)** | Chain reads | Apache-2.0 (installed package metadata) |
| **qrcode 8.2** | QR codes | BSD (installed package metadata; project licence file) |
| **MariaDB 10.11** | Database, unmodified | GPL-2.0 (GitHub repository licence); run as a separate service, not redistributed |
| **Redis 8.10 (local)** | Cache and queue, unmodified | Tri-licensed RSALv2, SSPLv1 or AGPLv3 from Redis 8. The hosted server's version was not confirmed. |
| **ODK Collect, ODK Central, pyxform** | Offline form capture and build | Apache-2.0, Apache-2.0, BSD-2-Clause (project licence files) |
| **Plus Jakarta Sans, JetBrains Mono** | Fonts bundled in the repository | SIL OFL 1.1 (project repositories). Licence text is not yet in the repository; add it when redistributing |
| **Lucide icons** | Icons inlined in templates | ISC, as stated in `templates/includes/icons.html` ; not independently checked |
| **jsonschema, Playwright** | Test tools only | MIT, Apache-2.0 (GitHub repository licences) |
| **GS1 EPCIS JSON schema** | Downloaded during tests | GS1 standard; fetched at test time, not redistributed |

| Component (version checked) | Role | Licence and how it was verified |
|---|---|---|
| **Stellar testnet and RPC** | Network service (API) | Public test network; no real funds |
| **Seed history dataset ( `axk_seed/` )** | Explorer history demo | Synthetic, created for this project and tagged as synthetic |

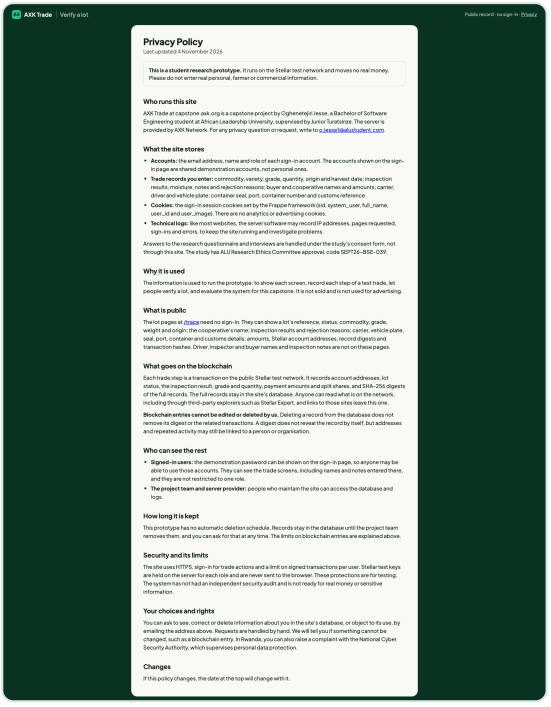
I will preserve copyright notices, identify third-party components, and avoid presenting community or vendor code as my own. Before any commercial redistribution, the inventory must be repeated against the deployed versions.

## 3.3 Privacy Policy Evidence and Implementation Gap

When this audit began, the application had no Privacy Policy or EULA: `/privacy` returned 404 on the live site on 4 November 2026. A policy page written to match the system's actual behaviour has been prepared and tested on the local stack, including its limits: shared logins, what is public, what is on-chain, and what cannot be deleted. As of 11 November 2026 it is not yet deployed: `/privacy` still returns 404 on capstone.axk.org.

**Figure 13**

*Privacy Policy Page (Local Preview)*




*Note.* Privacy Policy page on the local full stack, 4 November 2026. Local preview, not yet deployed.

## Discussion: Ethical Trade-Offs and Limits

The strongest ethical contribution of this Capstone is not a claim that blockchain creates trust. It is the attempt to make a chain of decisions visible: who inspected a lot, whether funds entered escrow only after approval, where settlement can go, and which evidence can be checked later. This audit also showed where the design falls short of its own language. Separation of duties holds at the contract level but not for signed-in users, while centralised attestations and server-held keys ask users to trust the operator. A trustworthy payment record is not the same as a trustworthy social process. My next priorities are an appeal path, per-role authorisation with

individual accounts, independent custody signatures, audited key management and measured access for low-resource users.

## References

African Leadership University Senate Research Ethics Committee. (2026, November 2). *Ethics approval for research project* (Approval code SEPT26-BSE-039) [Unpublished approval letter].

African Union. (2014). *African Union Convention on Cyber Security and Personal Data Protection*. https://au.int/en/treaties/african-union-convention-cyber-security-and-personal-data-protection

Association for Computing Machinery. (2018). *ACM code of ethics and professional conduct*. https://www.acm.org/code-of-ethics

East African Community & United Nations Conference on Trade and Development. (2008). *Draft EAC legal framework for cyberlaws*. https://www.tralac.org/documents/resources/eac/1386-draft-eac-framework-for-cyberlaws-november-2008.html

European Parliament & Council of the European Union. (2016). Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation). *Official Journal of the European Union, L 119*, 1–88. https://eur-lex.europa.eu/eli/reg/2016/679/oj

International Organization for Standardization & International Electrotechnical Commission. (2022). *Information security, cybersecurity and privacy protection: Information security management systems: Requirements* (ISO/IEC Standard No. 27001:2022). https://www.iso.org/standard/27001

Jesse, O. (2026a). *Architecture* [Software documentation]. GitHub. https://github.com/tejirijesse/axk-open-source-capstone/blob/44629ba555ccdfcf7828a81c3405f5a1202d2891/docs/architecture.md

Jesse, O. (2026b). *AXK open-source capstone* [Computer software]. GitHub. https://github.com/tejirijesse/axk-open-source-capstone

Jesse, O. (2026c). *Consent for participation: AXK Network prototype evaluation* [Unpublished consent form].

Republic of Kenya. (2019). *Data Protection Act, No. 24 of 2019* (Cap. 411C). https://kenyalaw.org/akn/ke/act/2019/24/eng@2022-12-31

Republic of Rwanda. (2021, October 15). *Law No. 058/2021 of 13/10/2021 relating to the protection of personal data and privacy*. Official Gazette n° Special of 15/10/2021. https://www.gov.rw/fileadmin/user_upload/RISA/Publications/2.Laws/Law_relating_to_the_protection_of_personal_data_and_privacy.pdf

Republic of Uganda. (2019). *Data Protection and Privacy Act, 2019* (Act 9 of 2019). https://media.ulii.org/media/legislation/18002/source_file/98f959926ba2429c/2019-9.pdf

# Appendix A: Technical Evidence Register

All code locations are at commit `44629ba`. Links open the exact file: https://github.com/tejirijesse/axk-open-source-capstone/blob/44629ba555ccdfcf7828a81c3405f5a1202d2891/

**8** implemented | **3** partial or by design | **5** not built or pending

**Table 8**

*Technical Evidence Register*

| Finding | Location | Status | Ethical significance |
|---|---|---|---|
| **Payment only for an inspected lot; no payee argument** | `contracts/escrow/src/lib.rs` 143-175 | Implemented, tested | Protects buyer funds and stops redirection |
| **Allow-listed, one-time inspection; owner refused** | `contracts/lot_registry/src/lib.rs` 198-226 | Implemented, tested | Separation of duties at contract level |
| **Rejection needs a listed reason** | `trade_api.py` 226 | Implemented | Decisions can be explained and compared |
| **Records hashed; digests anchored** | `ledger/records.py` 14-45 | Implemented, tested | Tamper evidence without publishing records |
| **Fixed public field list** | `verify.py` 16-26 | Implemented | Data minimisation; plate and customs still public |
| **Grade, quantity, result on-chain** | `contracts/lot_registry/src/lib.rs` 232-243 | By design | Permanent public commercial data |
| **Sign-in and rate limit** | `trade_api.py` 55-75 | Partial | Blocks guests, but no per-role check |
| **Demo password can appear on sign-in** | `demo.py` 21-37; live /login | By configuration | Signed-in screens are effectively public |
| **Failed chain call fails the step** | `trade_api.py` 78-84 | Implemented | No false records of completed steps |
| **Offline inspection import** | `odk.py` 69-101 | Implemented, tested | Reduces connectivity barrier for inspectors |
| **Refund after deadline without a signature** | `docs/security.md` guarantee 8 | Implemented, tested | Buyer is not dependent on AXK |
| **Inspection appeal or correction** | None; buyer disputes are ROADMAP BUY-4 | Not implemented | No remedy for a mistaken decision |
| **Self-signing by carrier and port** | ROADMAP LOG-4, PORT-1 | Not implemented | Records prove what the app recorded, not agreement |
| **Translations** | ROADMAP I18N-1 | Not implemented | Language exclusion remains |
| **Contract audit** | `docs/security.md` known limits | Not done | Not suitable for real funds |
| **Privacy Policy page** | Prepared; not deployed | Pending | Required before the video |